Figure 6.8. In tournament selection we evaluate the fitness of all the individuals in the population as usual, and then we choose a random subset of the full population (in this figure just 2 of 4), and then choose the top individuals (usually 2) in this subset, mate them to produce offspring and mutate them. We repeat this selection process until we fill up the next generation.

Figure 6.9. The average score of the population over generations in a genetic algorithm used to train agents to play CartPole.

Figure 6.10. In an evolutionary strategy we create a population of individuals by repeatedly adding a small amount of random noise to a parent individual to generate multiple variants of the parent. We then assign fitness scores to each variant by testing them in the environment, and then we get a new parent by taking a weighted sum of all the variants.

Figure 6.11. In an evolutionary strategy, at each time step we get an updated parameter vector by taking the old parameter vector and adding it to a weighted sum of the noise vectors, where the weights are proportional to the fitness scores.

Figure 6.12. Determining the fitness of an agent is often the slowest step in a training loop and requires that we run the agent through the environment (possibly many times). If we are doing this on a single computer, we will be doing this in serial—we have to wait for one to finish running through the environment before we can start determining the fitness of the second agent. The time it takes to run this algorithm is a function of the number of agents and the time it takes to run through the environment for a single agent.

Figure 6.13. If we have multiple machines at our disposal, we can determine the fitness of each agent on its own machine in parallel with each other. We do not have to wait for one agent to finish running through the environment before starting the next one. This will provide a huge speed up if we are training agents with a long episode length. You can see now that this algorithm is only a function of the time it takes to assess the fitness of a single agent, and not the number of agents we are assessing.